Portfolio Construction Workbook

Step-by-step solutions you can follow with a calculator

By hand

↓

No-code lab

↓

AI prompt

Financial ML · Week 11 Workbook

Correlation alone moves portfolio volatility from 15% to 5%

Vol 20% and 10%, 50% each — write 0.20, not 20

Step 1 · Corr. 0

$$\sigma_p^2 = 0.5^2(0.20)^2 + 0.5^2(0.10)^2 + 0 = 0.0100 + 0.0025 = 0.0125, \qquad \sigma_p = \sqrt{0.0125} \approx 11.18\%$$

Step 2 · Corr. 1

$$\rho = 1: \ \sigma_p^2 = 0.0100 + 0.0025 + 2(0.5)(0.5)(1)(0.2)(0.1) = 0.0225, \qquad \sigma_p = 15\%$$

Step 3 · Corr. −1

$$\rho = -1: \ \sigma_p^2 = 0.0125 - 0.0100 = 0.0025, \qquad \sigma_p = 5\%$$

Check you took the square root of variance

Inverse-variance and inverse-vol weight the same assets differently

Two assets, vol 20% and 10% — variances 0.04 and 0.01

Step 1 · Inverse-variance: ∝ 1/variance

$$\frac{1/0.04}{1/0.04 + 1/0.01} = \frac{25}{125} = 20\%, \qquad \frac{100}{125} = 80\%$$

Step 2 · Inverse-volatility: ∝ 1/vol

Inverse-volatility: $\frac{1/0.2}{1/0.2 + 1/0.1} = \frac{5}{15} = 33.3\%, \qquad \frac{10}{15} = 66.7\%$

Inverse-variance 20/80% and inverse-vol 33.3/66.7% differ

HRP first finds cluster variances, then splits inversely

Source tree, p. 18 — each variance 1, 50:50 in clusters

Step 1 · Defensive cluster

defensive cluster : $0.5^2(1) + 0.5^2(1) + 2(0.5)(0.5)(0) = 0.5$

Step 2 · Tech cluster

tech cluster : $0.5^2(1) + 0.5^2(1) + 2(0.5)(0.5)(0.8) = 0.9$

Step 3 · Across clusters

$$w_{\text{def}} = \frac{1/0.5}{1/0.5 + 1/0.9} = 64.3\%, \qquad w_{\text{tech}} = 35.7\%$$

Step 4 · Half each inside

$64.3\% \div 2 \approx 32.1\%, \qquad 35.7\% \div 2 \approx 17.9\%$

Compute two-asset risk to compare allocation methods

Calculator plus AI chat

1. Where

Use a calculator.

2. Input

Enter vol 20%, 10%, 50% each, correlation 0.

3. Output

You get variance 0.0125, vol about 11.18%.

4. Check

Set correlation to 1; expect 15%.

Prompt for the AI chat

Explain 1/N, inverse-variance, HRP and NCO to a non-major as team building. Hand-calculate equal-weight risk of two assets (vol 20%, 10%) at correlation 1, 0, −1. Compare inverse-variance and inverse-vol weights. Propose a table for return, risk, turnover and costs.

CALCULATE BY HAND.

CHECK THE DENOMINATOR.

THEN TRUST THE NUMBER.

Calculate once by hand
and the assumptions show.